Suraj Kiron Nair

Curriculum Vitae

Date of birth	17th May 1999	**Nationality**	Indian
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Personal Email	surajkiron1999@gmail.com	**Website**	surajkiron.github.io

EDUCATION

Aug 2017-July 2021

Bachelor of Engineering (B.Eng) Mechanical Engineering

Ramaiah Institute of Technology

- Graduated: First Class (1st) with Distinction
- GPA: 8.48/10

RESEARCH EXPERIENCE

Sept 2021-Present

Indian Institute of Science(IISc):
Interdisciplinary Center for Energy Research(ICER)
Thermal Systems Lab: Project Assistant (Research)
Advisor: Dr Pramod Kumar

My research at the Thermal systems Lab has been focused on the computational modelling of Printed Circuit Heat Exchangers .

- Worked on the 1D modelling of Printed Circuit Heat Exchangers (PCHEs) using Thermal Resistance Networks.
- Optimization of PCHEs using numerical methods.
- Structural analysis for Supercritical CO_2 Turbomachinery

Sept 2020 -Jan 2021

Indian Institute of Science(IISc):
Interdisciplinary Center for Energy Research(ICER)
Research Internship
Advisor: Dr Pramod Kumar

- 1D Modelling of Supercritical CO_2 Brayton cycle using Python.
- 3D transient simulation of Supercritical CO_2 reciprocating compressor using Ansys ICE module.

PROJECT EXPERIENCE

Feb 2021 -July 2021

Ramaiah Institute of Technology
Dept of Mechanical Engineering
Advisors: Asst Prof. Dr Jyothilakshmi and
Asst Prof. Dr Venkatesh Donekal

- Mini Project
  Cooling system design for 32650 cylindrical Li-ion Battery. This project involved thermal modelling of cells to determine their transient characteristics at varying discharge rates.
- Final Year Project:
  Robotic manipulator for sorting application based on colour detection. The Arduino controlled robotic arm is used sort colored blocks into seperate bins using a color sensor.

Aug 2017 -Jan 2021

Formula Student Electric Car
Ramaiah Institute of Technology: Dept of Mechanical Engineering
Team Vice-Captain and Powertrain Department Lead
Advisor: Prof. Gururaj Lalgi

- 5th place in Formula Bharat 2020 and Winner of Formula Green 2020
- 10th place Formula Bharat 2019 and Runner up Formula Green 2019
- Developed a battery pack and Powertrain system for a formula style electric vehicle.
- Conducted simulations to optimize lap times and longitudinal dynamics of the electric car using MATLAB and Simulink.
- Modeled battery performance characteristics and thermal behavior to determine required energy capacity and cooling system properties.

June 2019 -July 2019

Fenwick and Ravi: Design Department
Undergraduate Internship
Advisors: Mr Anil Varghese

Studied the manufacturing processes involved in development of Hydraulic Actuators.
Designed, manufactured and tested Hydraulic Cylinders.

ACTIVITIES

Aug 2017-Jan 2019

National Service Scheme(NSS)

Volunteer Blood Donation Camp (Event photographer 2018,2019), Polio vaccine drive(2018) and Orphanage Support program. Volunteer 9th ICCCNT (IEEE) Conference

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RESEARCH INTERESTS

My primary interests are non-linear control systems, dynamics and Robotics. I am keen on developing navigation, and computer vision based control methods that can be implemented on automobiles and robotics. Moreover, I am interested in simulating such systems using computational/numerical methods and developing systems to integrate this technology with electric vehicles.

COURSES AND CERTIFICATIONS

Solidworks: Certified Associate Mechanical Design (CSWA)
Programming Numerical Methods on Python (Udemy)
ROS for beginners (Udemy)
Vocational course in IOT: Ramaiah Institute of Technology
Machine Learning: Stanford Online
MATLAB Onramp and Simulink Onramp
Deep Learning Specialization: Deeplearning.ai

SKILLS

CAD:
Solidworks, Creo, AutoCAD

Manufacturing:
Additive Manufacturing, CNC

Programming:
C, C++, MatLab, Arduino
Python, Tensorflow, ROS

Simulation:
Ansys (Fluent and structural),
Flownex , Solidworks simulation.